		Since speed is maximum at equilibrium position, mass has maximum speed at time t = T/4.
16	**B**	Total energy $E = \frac{1}{2} m\omega^2 A^2$ $E_K = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right)$ $\frac{3}{4} E = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right)$ $\frac{3}{4}\left(\frac{1}{2} m\omega^2 A^2\right) = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right)$ $x = \frac{A}{2}$
17	**A**	B: There is no displacement of a particle at a **node** at any time. C: **Half a** wavelength of the stationary wave is equal to the separation of two